Formal Abductive Latent Explanations for Prototype-Based Networks

Context

Existing eXplainable AI (XAI) methods tend to generate explanations **not faithful** to the decision process.

Two distinct solutions have been previously proposed:

- *interpretable-by-design* models, like **ProtoPNet**
  - ☺ rely on human-readable concepts
  - ☹ **optimistic** explanations (insufficient)
- formal abductive explanations **AXps**
  - ☺ guarantee the prediction (sufficient)
  - ☹ only at the pixel level

Our solution leverages both approaches to generate *interpretable* **Abductive Latent Explanations** (ALE).

Formalism

$$\forall \mathbf{x} \in \mathcal{F}. \quad \phi_{\mathcal{E}}(f(\mathbf{x}), f(\mathbf{v})) \Rightarrow (\kappa(\mathbf{x}) = \kappa(\mathbf{v})).$$

Intuition: in the case of ProtoPNet-*like* neural architectures, an ALE of $\mathbf{v}$ is a subset $\mathcal{E}$ of distances between learned *prototypes* and latent representation components $f(\mathbf{v})$, such that the omitted distances, which are approximated, guarantee the prediction $\kappa(\mathbf{v})$.

Types of ALE

- **Top-*k***
  output the *k* most activated prototypes
- **Spatial Approximations**
  - Triangle Inequality
  - Hypersphere Intersection

Motivating Example

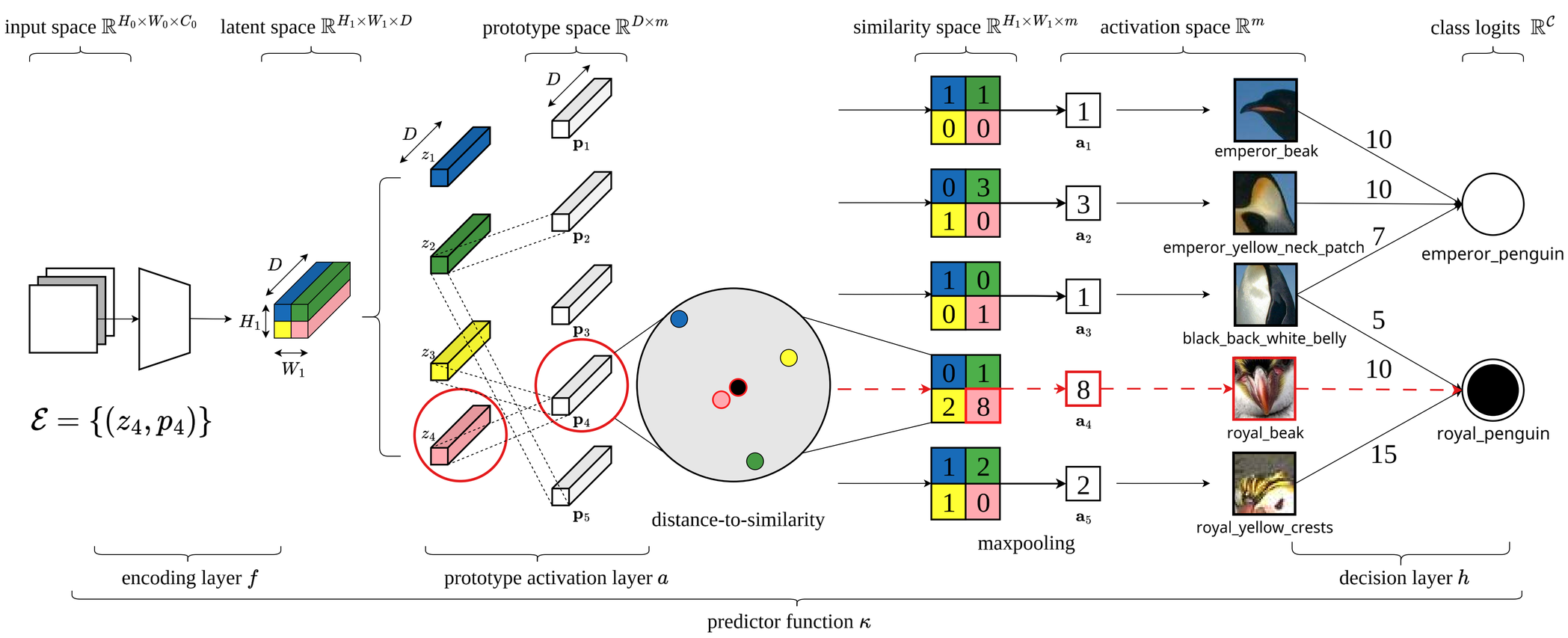


Fig 1. ProtoPNet top-1 explanation for the decision « royal penguin »

Results

- Existing models in the literature produce *insufficient* explanations
- Yielding *sufficient* explanations requires adding **many** prototypes
- When the latent resolution is low, spatial ALEs are smaller than Top-*k* explanations—when the latent resolution is high, spatial ALEs are (mechanically) higher than Top-*k* explanations.
- When the model is right, the explanation is smaller than when the model is wrong.

Future Works

- Extending the formalism and spatial approximations to other functions of distance or similarity : cosine similarity, dimension projection, focal similarity, gaussian distance, etc.
- Remove background patches from the explanation.
- Use the formal explanations, and the generation process, to increase the trustworthiness and the predictive performance of the model.
- Create new prototype-based networks, with richer similarity.

Contact : Jules Soria / jules.soria@cea.fr